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PI Name:	Grulke, Eric Ph.D.		
Project Title:	Synthesis and Analysis of Nanoparticle Composites for Space Radiation Shielding		
Division Name:	Physical Sciences		
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Program/Discipline--Element/Subdiscipline:	RADIATION SHIELDING--Radiation shielding		
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Comments:			
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No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
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Contact Monitor:		**Contact Phone:**	
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COI Name (Institution):	Hertel, Nolan E. (Georgia Tech) Bertelsen, Craig (Adhesion Solutions) Weaver, Sean (Adhesion Solutions)		
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Task Description:	This project will synthesis nanoparticle composites suitable for radiation shielding, and analyze their radiation and structural properties. Our objective is to develop nanocomposites with the following multifunctional characteristics: 1) radiation shielding near that of polyethylene, 2) tensile and flexural strength near that of polyethylene, and much higher toughness and impact strength than that of polyethylene, and 3) excellent flame retardancy. We also propose determining the ballistic and/or hypervelocity impact performance of our nanocomposites. The research team is materials engineer with experience in nanoparticles dispersions and nanocomposites (University of Kentucky), a nuclear engineer with extensive experience in radiation shielding measurements (Georgia Institute of Technology), and a small business specializing in adhesion and interfacial chemistry (Adhesion Solutions, Inc.). Composites made by dispersing nanoparticles into polymers can have improved mechanical and transport properties. Nanoscale dispersions of asymmetric nanoparticles (long cylinders or flat disks) with high thermal conductivities can give polymers with improved flame retardancies, and may improve impact performance at very high strain rates. We will use components with excellent space radiation shielding properties: polyethylene as the continuous phase and boron-containing nanoparticles as the dispersed phase. While polyethylene/boron blends have been previously considered for radiation shielding, our focus will be to synthesize nanophase dispersions (< 100 nm in discontinuous phase size). High interfacial adhesion between the polymer and nanoparticles phases will be needed to obtain the best multifunctional material performance. We will use thin film adhesion testing methods to accelerate research on surface functionalization methods. Radiation shielding tests will demonstrate whether there are differences in radiation shielding based on nanoparticle type, orientation, and dispersion quality. These multifunctional nanocomposites could be fabricated using conventional polymer process technologies, and would be suitable for use in composite sandwich constructions, as foams and coatings, as filled fibers, and as structural polymers in a variety of spacecraft applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Space radiation shielding is an important concern to scientists as they prepare for long term missions to protect the astronauts from fatal doses of radiation. Polyethylene is already a widely used shielding material because of it as low cost and availability. Composites containing polyethylene and a boron-rich filler are thought to increase the shielding ability over neat polyethylene. Boron is effective at shielding radiaiton by capturing neutrons created by secondary reactions. It has been shown adding boron to polymeric materilas allow it to absorb low-energy neutrons while not altering the shielding ability of the composite to stop high energy heavy ions. Another benefit of using polyethylene composites as radiation shields is their ability to be structural components. Boron-rich composites are advantageous because of their ability to be a radiation shielding material and also their increased mechanical properties.
Task Progress:	Composites based on two boron-rich materials, boron nitride and boron carbide, were synthesized and analyzed for particle loadings up to 15 volume %. These materials were examined under an optical microscope and scanning electron microscope and found to be randomly distributed. These inorganic compounds are able to disperse and not agglomerate in a polyethylene matrix. Mechanical properties of these composites show an increase in modulus for boron nitride and a larger increase for boron carbide. Tensile strength was unaffected by the addition of a boron nitride filler. The addition of filler to the polyethylene matrix, however, did decrease the toughness for both of these composites. The filler acts as a void in the matrix, and causes the composites to fail at a lower strain values. Surface functionalization methods were used to improve adhesion between the particle and the matrix. Before these surface modified fillers were introduced into the polyethylene matrix, the powders were examined using FT-IR. This tool was used to verify if the surface modification was successful by looking for the silane group on the surface. For each of the coupling agents used, a change in the surface structure is easily seen with the addition of a Si peak. A set of silane coupling agents were found to provide good adhesion between the inorganic particles and high density polyethylene. Adding fillers to ductile polymers such as polyethylene often results in lower mechanical properties and in particular, losses in toughness. Composites in which the fillers were functionalized with preferred silane coupling agents had higher moduli than those based on unfunctionalized particles. These treatments also showed slightly improved tensile strengths than the untreated particles. One of the functionalization systems made composites with higher toughness than the unfunctionalized controls. It seems likely that further work on the functionalization methods could lead to higher moduli, better tensile and toughness values. Mechanical properties are closely related to interfacial adhesion between the particles and the polymer matrix. Poor adhesion causes reduced load transfer and mechanical properties adversely affected. Without surface modification there is a weak adhesion present that leads to inadequate stress transfer and failure. With coupling agents, a direct contact was made between particles and matrix causing the composite to fail due to shear yield and tearing rather than voids present. It has been reported that coupling agents improve dispersion and also cause strong increases in yield strength and modulus. Our results show an increase in modulus and tensile strength for particles treated with coupling agents. But with an increase in volume percent filler, a decrease in tensile strength is seen. Morphologies at fracture surfaces provide information about the failure mechanisms and adhesion between particles and the matrix. Looking at the surface of the composite without surface treatments shows very poor adhesion. There are holes on the fractured surface where it appears the particle has pulled out of the matrix. Where the particle remains in the matrix, an interface containing only air surrounds the particle. With poor interfacial adhesion present, this composite failed by the particle acting as a void and providing no stress transfer to the matrix. Examining the surfaces of the composites where the particles had been modified shows very good adhesion. From 1 to 15 volume percent filler the number of particles increases and continues to appear to be dispersed. Particles can still be seen in each of the holes on the surface and did not debond from the matrix. Looking closely at the interface shows the particles are connected to the matrix and good adhesion is achieved. All of the coupling agents show improved adhesion and therefore agree with the mechanical properties observed. Three composites, PE/BN powder, PE BN nanoparticles and PE/B4C powder, were tested for radiation shielding effects in comparison with aluminum and polyethylene as controls. Experimental results from both the WNR and Fermilab tests showed that aluminum, the material conventionally used for space shielding, exhibited the most effective shielding against high-energy protons and the worst shielding properties for neutrons of the five materials (Al, PE, and three composites). Although the polyethylene-based materials did exhibit similar shielding behavior, pure polyethylene was the least effective shielding material for protons. Boron nitride and nano-boron nitride displayed equivalent shielding

	properties as polyethylene for high-energy neutrons. Boron nitride, nano-boron nitride, and boron carbide displayed approximately equivalent shielding properties for high energy protons. Boron carbide proved to be the most effective shielding material for high energy neutrons. Overall, boron carbide was the best overall shielding material for high energy protons and neutrons. The goal of this project was to develop a nanoparticle composite suitable for radiation shielding and analyze the radiation and structural properties. The composite was intended to have multifunctional characteristics that include shielding near that of polyethylene and mechanical properties superior to polyethylene. Through this work we have optimized the mechanical properties for micron size boron nitride particles with surface treatments. These treatments can be applied to boron carbide as well as the nanoparticles of boron nitride and boron carbide and expect to show similar results. Testing for high energy photons and also neutrons shows the boron nitride composites have shielding characteristics near that of polyethylene. Boron carbide is the best material of those tested for shielding high energy photons and neutrons. These materials tested for radiation shielding properties have no surface treatments to enhance their mechanical properties. It is thought that testing the materials with improved adhesion, and hence mechanical properties, would provide shielding properties better than polyethylene. A consistent set of functionalization tools has been developed through this work; (1) silane coupling agent selected, (2) reactivity determined, (3) morphology of melt blend examined, (4) mechanical properties tested, and (5) radiation shielding potential determined. This suite of tools has enabled us to accelerate the study of radiation shielding materials. This work will continue by looking at surface functionalization of boron carbide particles. Mechanical properties and radiation shielding data have shown boron carbide composites to be superior and modifying the surface should improve these properties. Nanoparticles will also be used and surface modifications made to optimize their composites mechanical and radiation shielding properties. Additional testing will be done to determine shielding effectiveness at Georgia Tech and also at Penn State University.
Bibliography Type:	Description: (Last Updated: 04/09/2007)
Abstracts for Journals and Proceedings	Shannon M, Burgett E, Hertel N, Burns K, Blaylock D. "Nanoparticle material evaluation as a space radiation shielding material." 2007 ANS Annual Meeting, Boston, MA. June 24-28, 2007. Transactions. 2007 ANS Annual Meeting, June 2007. , Jun-2007
Abstracts for Journals and Proceedings	Shannon M, Burgett E, Hertel N, Blaylock D, Bruns K, Dewji S, Lobracco C, Howell R, Harrison C, Grulke E. "Testing of a novel shielding material in high energy particle beams." 52nd Annual Meeting of the Health Physics Society, Portland, OR, July 8-12, 2007. 52nd Annual Meeting of the Health Physics Society, July, 2007. , Jul-2007
Articles in Peer-reviewed Journals	Harrison C, Weaver S, Bertelson C, Grulke E. " Boron nitride composites for space radiation shielding. Mechanical properties." in preparation for Polymer. March 2007. , Mar-2007